Ethan Moon

616-295-5694 | ethmoon@umich.edu | linkedin.com/in/ethan-moon-b9a2a7314 | github.com/emoon0108 | emoon0108.github.io

EDUCATION

B.S.E. Computer Science, College of Engineering — University of Michigan | Ann Arbor, MI **Expected May 2028**

Coursework: Computers & Programming, Discrete Math, Proof-Based Linear Algebra, Multivariable Calculus, Differential Equations, Statistics, Physics

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL | Product: React, Next.js, React Native/Expo, Node.js, Hono | Data & Platform: PostgreSQL, Drizzle ORM, OpenAPI/Orval, Docker, Git | Engineering: OpenFOAM, ParaView, NASA GMAT, Arduino, ESP32, computer vision, digital twins | Spoken: English and Korean (native/bilingual), Spanish (limited working)

EXPERIENCE

Lead Engineer & Head of U.S. Team — Ody **May 2026 – Present**

- Shipped 107 merged PRs in 10 weeks across six product apps and the backend, owning features from PostgreSQL/Drizzle schemas and Hono APIs through OpenAPI/Orval clients to React and Expo interfaces.
- Delivered four merchant-facing platform features, including AI-assisted multilingual digital menus, email campaigns, and multi-location analytics, as secure production workflows.

Co-Author Researcher — Grand Valley State University **Sep 2025 – Present**

- Developed and calibrated non-Newtonian, multiphase CFD models in OpenFOAM/RheoTool across 32 bioink-deposition conditions, achieving 96% agreement with experimental data.
- Engineered a Dockerized Python/JavaScript analysis platform with VTK-to-CSV pipelines and interactive visualizations for 15,300+ records across 51 timesteps per simulation.
- Built real-time nozzle-anomaly detection using computer vision and digital-twin modeling; co-authored a peer-reviewed ASEE paper and a bioprinting journal manuscript in preparation.

Summer Researcher — Grand Valley State University **Jun 2025 – Aug 2026**

- Modeled orbital transfers, rendezvous, perturbations, and fuel-efficient formation flight in Octave and NASA GMAT; separately designed 3D-printed airfoils and Arduino-instrumented wind-tunnel experiments to study aerodynamic hysteresis.

Simulation & Robotics Engineering Intern — Casual Simulation **Feb 2024 – Jan 2026**

- Built a VR rowing-performance simulation adopted by a local high-school team and improved Arduino-based robotic-dog controls, increasing stability by 21%.

SELECTED PROJECTS

No Spoilers — Full-Stack Web App & Chrome Extension — nospoilers.xyz **2026 – Present**

- Built spoiler-free discovery and watchlist workflows spanning 120 API handlers, 43 PostgreSQL models, and 42 user-facing page entry points; the privacy-first extension classifies spoilers on-device and the repository passes 54 automated tests.

ViolinTwin — AI Practice Coach — React Native/Expo **2026**

- Prototyped real-time practice feedback with native Swift/Kotlin analysis scaffolding, audio capture, pitch/rhythm/note-timing estimation, and MusicXML/MIDI targets.

Smart Plate — Accessible Nutrition App — Defold/Lua **2023 – 2025**

- Designed low-stimulation, gamified food-exposure workflows and a BLE-ready smart-plate architecture; a real-world pilot increased willingness to try new foods.

Founder & Lead Developer — Helping Hands Soccer Training **2022 – 2026**

- Built web/mobile matching and progress-tracking tools for a volunteer coaching nonprofit that donated $2,400 in scholarships.

HONORS

DECA State Champion | 2nd Place, GYM International STEM Innovation Challenge | International Astronomy & Astrophysics Competition Finalist | Dartmouth Book Award | Published, International STEM Research Journal